

Architecture Document

Integrated Performance Management System (IPMS)

3.1 System Overview

The Integrated Performance Management System (IPMS) is a web-based information system that automates and facilitates the management of the University of Antique employee performance planning, monitoring, evaluation, approval workflow, document tracking, and performance reporting. The system was developed to overcome the weaknesses of the existing manual, paper-based performance management system which is characterized by heavy reliance on printed Individual Performance Commitment and Review (IPCR) and Office Performance Commitment and Review (OPCR) forms, physical movement of documents, and manual updating of records. The traditional methods are often characterized by late submissions and approvals, lost or damaged documents, irregular tracking of performance records, and inefficient communication between faculty, staff, supervisors and administrators.

The proposed platform offers a common environment for employees, supervisors, department heads, HR admins and system admins to perform their jobs through a web browser using desktop or mobile based devices. Employees can set and submit performance objectives, update accomplishments, upload supporting documentation, track evaluation progress, and receive notifications on approvals, deadlines and evaluation schedules. Department heads and supervisors can view submitted objectives, evaluate employee performance, make comments and ratings, and approve or return submissions electronically. Human Resource administrators will be able to monitor the entire performance management process, generate reports at the institution level, track the progress of evaluations, manage user accounts and ensure that the university's performance management policies are followed.

The system consists of several major components. The frontend application is the user interface for interacting with the platform. The backend application was developed using the Laravel Framework. This application is responsible for handling user requests, enforcing business rules, calculating performance ratings, managing approval workflows, generating analytics and reports, and communicating with external services. The system data including user accounts, employee profiles, performance objectives, KPI records, evaluation results, approval records, uploaded supporting documents, audit logs, notifications and generated reports are stored in the MySQL database.

The platform has automated notification services to give instant updates on submission deadlines, approval decisions, evaluation schedules and other key performance management activities to boost efficiency and transparency. The system also includes audit logging for the recording of important user actions, document tracking for tracking the progress of performance documents throughout the evaluation workflow, and AI assisted recommendation functions, which provide performance insights and evaluation recommendations to support supervisors during the evaluation process.

Everything considered, the Integrated Performance Management System (IPMS) is a complete solution that boosts productivity, increases accountability and transparency, fortifies performance monitoring, and facilitates well-informed decision-making through automated workflows, centralized data management, and sophisticated reporting features.

3.2 Integration Pattern

The Integrated Performance Management System follows a Hub-and-Spoke Integration Pattern, where the Laravel Backend API serves as the central hub coordinating communication among all system components. The React frontend, MySQL database, AI recommendation service, notification service, and document tracking module communicate exclusively through the backend API. This architecture centralizes business rules, enhances security, simplifies maintenance, and supports future system enhancements.

The system also incorporates Event-Driven Architecture to automate important workflows. Examples include:

- Submission of an IPCR/OPCR automatically notifies the assigned supervisor.
- Approval or rejection automatically updates employee status.
- Upcoming deadlines automatically trigger reminder notifications.
- Completion of evaluations automatically generates reports and analytics.
- Completion of performance reviews automatically updates historical performance records.
- AI recommendations are generated after evaluation completion.

3.3 Data Contracts

Contract 1 – User Login

Endpoint

POST /api/login

Request

Field	Type	Required
email	String	Yes
password	String	Yes

```
{  
  "status": "success",  
  "token": "jwt_token",  
  "user": {  
    "id": 1,  
    "name": "Juan Dela Cruz",  
    "role": "Supervisor"  
  }  
}
```

Validation

- Valid email
- Password required

Contract 2 – Performance Objective Submission

Endpoint

POST /api/objectives

Field	Type	Required
objective	String	Yes
KPI	String	Yes
target	Integer	Yes
due_date	Dte	Yes
attachment	File	Optional

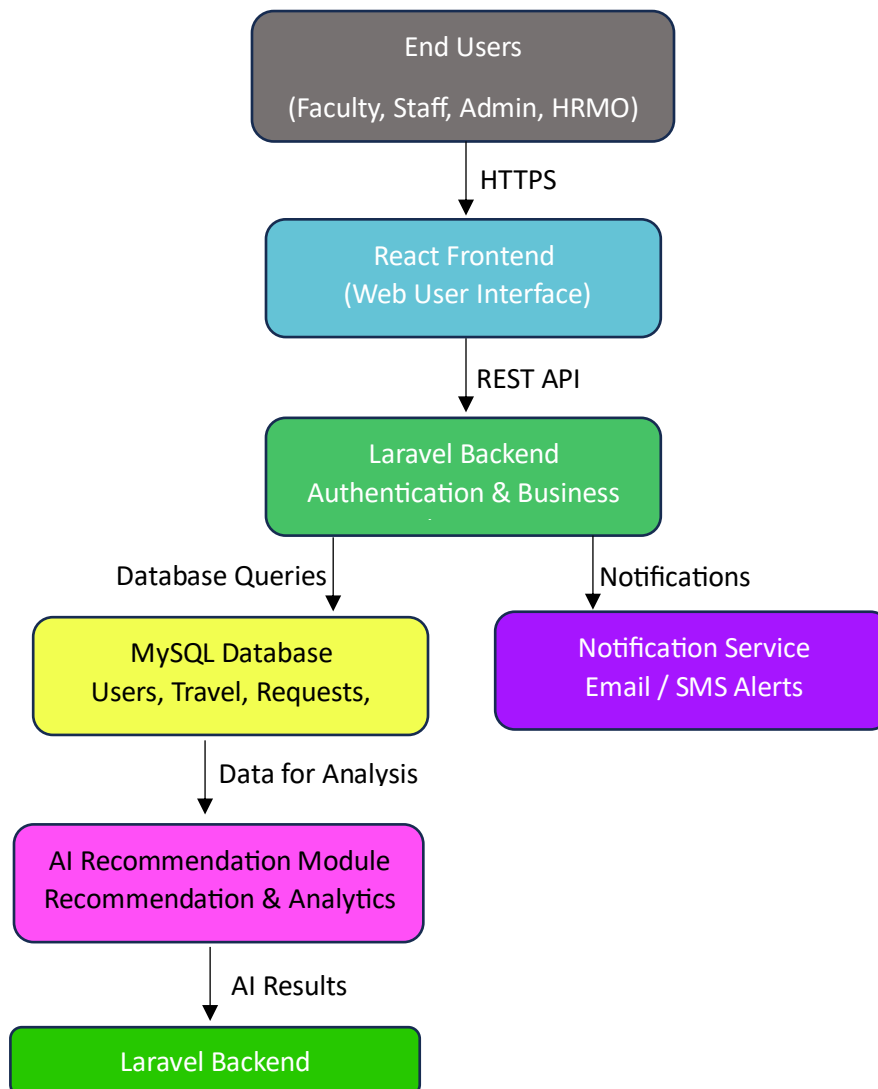
```
{  
  "status": "success",  
  "message": "Performance objective submitted successfully."  
}
```

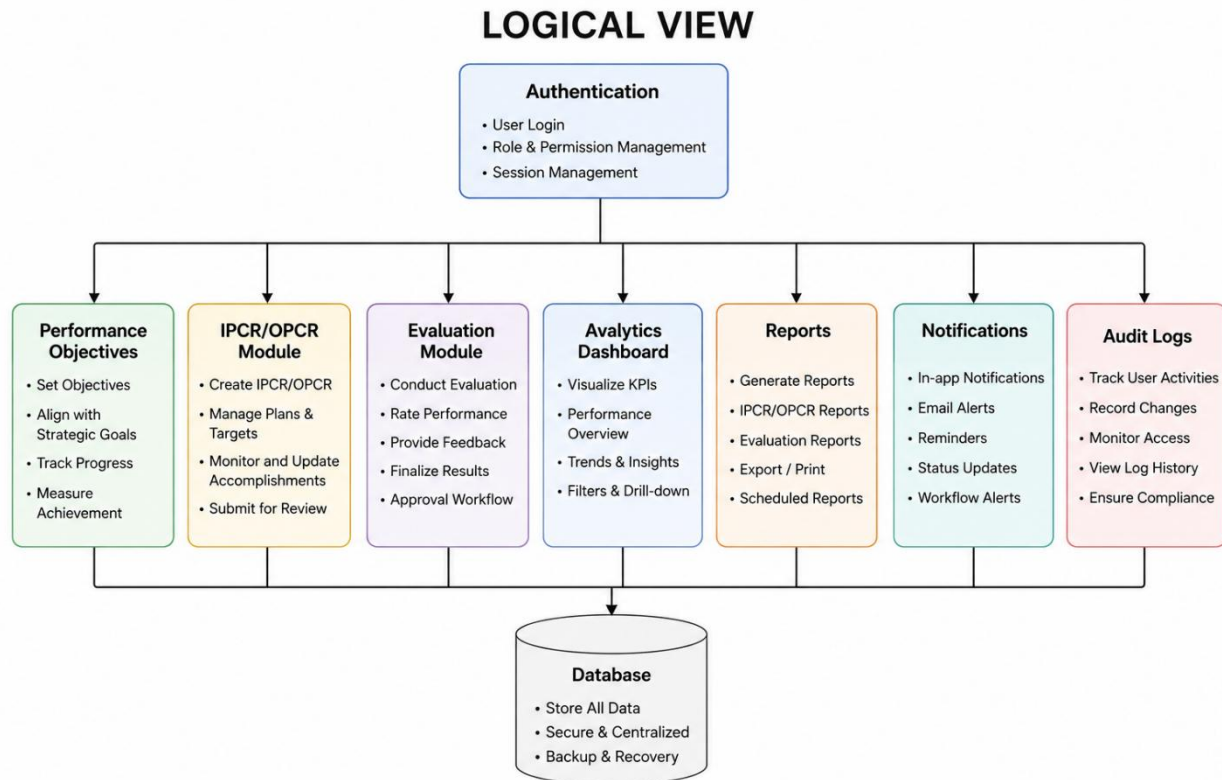
Validation

- Objective required
- KPI required
- Target must be numeric
- Due date required

3.4 Architectural Viewpoints

Deployment View





3.5 Security Considerations

- Authentication – JWT or Laravel Authentication for secure login.
- Authorization – Role-Based Access Control (Employee, Supervisor, HR Admin, System Administrator).
- Data Protection – MySQL database with encrypted passwords using bcrypt.
- Secure Communication – HTTPS.
- Document Security – Secure storage of IPCR, OPCR, and supporting documents.
- Audit Logging – Record login attempts, submissions, approvals, evaluations, and updates.
- Risk Assessment – Unauthorized access, data breaches, malware, accidental deletion, and service

3.6 Architectural Trade-offs

Laravel Monolithic vs Microservices

Advantages

- Easier deployment

- Faster development
- Lower infrastructure cost
- Suitable for a university capstone project

Disadvantages

- Less scalable than microservices
- Larger codebase as features grow

MySQL vs NoSQL

Advantages

- Strong relational database
- Reliable transactions
- Excellent reporting capability
- Ideal for structured performance records

Disadvantages

- Less flexible for unstructured data

AI-Assisted Evaluation vs Manual Recommendation

Advantages

- Faster evaluation support
- Consistent recommendations
- Reduced manual effort
- Improved decision-making

Disadvantages

- Recommendations still require human validation

Automated Notifications vs Manual Communication

Advantages

- Real-time reminders
- Faster approval workflow
- Reduced administrative workload

Disadvantages

- Depends on internet connectivity and email services